

TWITTER DATA ANALYTICS DASHBOARD

Internship Project Report

Project Type: Power BI Analytics Dashboard

Domain: Data Analytics

Dataset: Twitter Analytics Training Dataset

1. Project Overview

The Twitter Data Analytics Dashboard is an interactive Microsoft Power BI project developed as an extension of the original Twitter Analytics training project. The six internship tasks were implemented using the same training project and dataset, in accordance with the internship requirement to build on the existing project rather than use an unrelated dataset or new project.

2. Project Objectives

- Analyze tweet activity and engagement patterns.
- Compare interactions across tweet categories.
- Analyze media-related interactions across days of the week.
- Compare replies, retweets and likes over the specified period.
- Analyze monthly engagement rate trends.
- Identify top-performing tweets based on total engagement.
- Present the analysis through an interactive Power BI dashboard.

3. Dataset

The project uses the same Twitter Analytics training dataset used during the training project. It contains tweet-level fields including tweet date/time, impressions, engagements, engagement rate, retweets, replies, likes, URL clicks, user profile clicks, hashtag clicks, media views, media engagements, app activity, tweet category, word count and character count.

4. Data Preparation & Transformation

Data preparation and transformation were performed in Power Query within Power BI.

- Handled missing/null values.
- Removed duplicate records.
- Removed errors from date/time fields.
- Created tweet date, day, month, quarter and hour fields.
- Created word-count and character-count fields.
- Created even/odd date and impression conditions.
- Created tweet-category and media-status fields.
- Created calculated measures/columns required for the internship tasks.
- Created time-visibility logic for task visualizations.

5. Tools & Technologies

- Microsoft Power BI – dashboard development and visualization.
- Power Query – data cleaning and transformation.
- DAX – calculated measures and analytical logic.
- GitHub – project repository and documentation.

6. Dashboard Pages

- Home – project landing page.
- Navigation – navigation to the main dashboard sections.
- Key KPIs – overall performance indicators.
- Insights – concise analytical findings.
- Overview – overall tweet activity and engagement.
- Task 1 – Tweet Interaction Breakdown by Category.
- Task 2 – Engagement Rate Comparison.
- Task 3 – Media Interaction by Day of Week.
- Task 4 – Replies, Retweets and Likes Comparison.
- Task 5 – Monthly Engagement Rate Trend.
- Task 6 – Top 10 Tweets by Engagement.
- What is Twitter? – project context.
- Task Summary – summary of all six internship tasks.

7. Internship Task Implementation

Task 1 – Tweet Interaction Breakdown by Category

Requirement: Create a clustered bar chart showing URL clicks, user profile clicks and hashtag clicks grouped by tweet category, including tweets with at least one required interaction. The visual is visible only from 3 PM to 5 PM IST. Tweet date is filtered to even dates and word count is greater than 40.

Implementation: Implemented with the required task conditions. The visual may appear blank because the provided dataset has no matching records after the Word Count > 40 condition is applied together with the other required conditions.

Task 2 – Engagement Rate Comparison

Requirement: Compare engagement rate for tweets with app opens versus tweets without app opens. Include tweets posted between 9 AM and 5 PM on weekdays, use the required visibility windows, filter even impressions and odd tweet dates, require character count above 30, and exclude tweets containing the letter D.

Implementation: Implemented with the required filters and time-visibility logic. App Opens information is not sufficiently populated in the provided source data for the intended comparison, so this limitation is documented rather than relaxing the task requirements.

Task 3 – Media Interaction by Day of Week

Requirement: Create a dual-axis chart showing media views and media engagements by day of week for the last quarter. Apply the specified visibility windows, even impressions, odd tweet dates, character count above 30 and exclusion of tweets containing H.

Implementation: Implemented using the required filters, visibility logic and last-quarter selection. The selected quarter is Quarter 4.

Task 4 – Replies, Retweets and Likes Comparison

Requirement: Create a simple visualization comparing replies, retweets and likes for tweets posted from June through August 2020 using SUM aggregation.

Implementation: Implemented as the required engagement-metrics comparison for 1 June 2020 through 31 August 2020, inclusive.

Task 5 – Monthly Engagement Rate Trend

Requirement: Create a line chart showing average engagement rate by month, with separate lines for tweets with media and tweets without media.

Implementation: Implemented as a monthly trend comparison using average engagement rate and media status. No additional task-specific filters were applied.

Task 6 – Top 10 Tweets by Engagement

Requirement: Identify the Top 10 tweets based on the sum of retweets and likes, exclude weekends, apply the required visibility, even-impression, odd-date and word-count conditions, and display the associated user profile.

Implementation: Implemented with the required Top 10 engagement logic and filters. User profile information is not available in the provided dataset, so the requested profile field cannot be displayed.

8. Dashboard Outputs

The dashboard provides KPI-level and visual analysis of Twitter performance. The Key KPIs page summarizes overall engagement and interaction metrics. The Overview page presents tweet activity by time and overall engagement patterns. The task pages provide the specific analyses required by the internship.

9. Key Insights

- Tweet activity changes across posting hours and days.
- Different tweet categories generate different types and levels of interaction.
- Media views and media engagements vary by day of week.
- Average engagement rate changes across months.
- The Top 10 analysis highlights tweets with the highest combined retweet and like engagement.

10. Dataset Limitations

- Task 1: No matching records are available after applying the required Word Count > 40 condition together with the task filters.

- Task 2: App Opens data is not sufficiently populated in the provided source dataset for the intended comparison.
- Task 6: User profile information is not available in the provided dataset.

The required internship filters and conditions were retained. The dashboard does not relax task requirements simply to force a visual result.

11. Submission Resources & Project Links

Resource	Link
Hosted Power BI Dashboard	https://app.powerbi.com/links/RM0mvDx8o4?ctid=8e264576-b9fc-4c76-b018-2ff84125859b&pbi_source=linkShare
GitHub Repository	https://github.com/NandhanaKS-BI/twitter-analytics-dashboard
Google Drive Dataset	https://docs.google.com/spreadsheets/d/1jeGZX3lyBPfopkadfay-M2cgboR9MxrU/edit?usp=sharing&ouid=113351234268593457644&rtpof=true&sd=true
Project Report	Final project report

12. GitHub Repository Structure

- README.md – project overview, dashboard details, links and limitations.
- Dataset/ – project dataset.
- Documentation/ – final project report.
- Dashboard/ --Twitter_Analytics_Dashboard.pdf, Twitter_Analytics_Dashboard.pbix

13. Project Outcome

The completed dashboard demonstrates practical Power BI skills including data preparation, Power Query transformation, calculated logic, filtering, aggregation, KPI reporting, interactive visualization, navigation and task-specific analytics. All six internship requirements were implemented on the same Twitter Analytics project and dataset.

14. Conclusion

The Twitter Data Analytics Dashboard provides a concise and interactive view of tweet activity, engagement, interactions, media performance, trends and high-performing tweets. The project follows the internship requirement to build the assigned analytics tasks on the existing training project and dataset. Where the supplied dataset does not contain sufficient records or required fields, the limitation is documented transparently.